

Tyler Chen

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<https://research.chen.pw>

Positions

JPMorganChase (2025-present)

Vice President, Applied Research Lead

- Quantum-inspired and Randomized Algorithms team at Global Technology Applied Research, JPMorganChase

New York University (2022-2025)

Assistant Professor / Courant Instructor

- Mathematics at Courant, Computer Science and Engineering at Tandon
- Sponsor: Christopher Musco

University of Washington (2017-2021)

Academic Student Employee

- Instructor and Teaching Assistant (unionized with UAW Local 4121)

Education

University of Washington 2017-2022

Ph.D. in Applied Mathematics

- Thesis: *Lanczos-based methods for matrix functions*
- Advisors: Anne Greenbaum, Thomas Trogdon

University of Washington 2017-2019

M.Sc. in Applied Mathematics

Tufts University 2013-2017

B.S. Summa Cum Laude in Mathematics and Physics; Minor in Studio Art

Research Interests

I'm particularly interested in incorporating probabilistic techniques into classical algorithms to develop methods which are fast and reliable, both in theory and in practice. I hope that my work will help to bridge the gaps between numerical analysis, theoretical computer science, and applied computational sciences such as quantum physics with the ultimate goal of supporting the advancement of knowledge in the basic sciences.

Monographs

Randomized Numerical Linear Algebra with Examples

Tyler Chen. 2025. [url].

- Online book providing a first introduction to RandNLA, with many accompanying numerical examples.

The Lanczos algorithm for matrix functions: a handbook for scientists

Tyler Chen. 2024. [arXiv].

- Accessible introduction to Lanczos-based methods for matrix functions, with a particular emphasis on conceptual understanding of the algorithm in finite precision arithmetic.

Papers (in progress)

A simple analysis of a quantum-inspired algorithm for solving low-rank linear systems

Tyler Chen, Junhyung Lyle Kim, Archan Ray, Shouvanik Chakrabarti, Dylan Herman, and Niraj Kumar. 2025. [arXiv]

Query Efficient Structured Matrix Learning

Noah Amsel, Pratyush Avi, Tyler Chen, Feyza Duman Keles, Chinmay Hegde, Cameron Musco, Christopher Musco, and David Persson. 2025. [arXiv]

GPU-Parallelizable Randomized Sketch-and-Precondition for Linear Regression using Sparse Sign Sketches

Tyler Chen, Pradeep Niroula, Archan Ray, Pragna Subrahmanya, Marco Pistoia, and Niraj Kumar. 2025. [arXiv]

Quasi-optimal hierarchically semi-separable matrix approximation

Noah Amsel, Tyler Chen, Feyza Duman Keles, Diana Halikias, Cameron Musco, Christopher Musco, and David Persson. 2025. [arXiv]

Provably faster randomized and quantum algorithms for k -means clustering via uniform sampling

Tyler Chen, Archan Ray, Akshay Seshadri, Dylan Herman, Bao Bach, Pranav Deshpande, Abhishek Som, Niraj Kumar, and Marco Pistoia. 2025. [arXiv]

Preconditioning without a preconditioner: faster ridge-regression and Gaussian sampling with randomized block Krylov subspace methods

Tyler Chen, Caroline Huber, Ethan Lin, and Hajar Zaid. 2025. [arXiv]

Randomized block-Krylov subspace methods for low-rank approximation of matrix functions

David Persson, Tyler Chen, and Christopher Musco. 2025. [arXiv]

Papers (published)

Does block size matter in randomized block Krylov low-rank approximation?

Tyler Chen, Ethan N. Epperly, Raphael A. Meyer, Christopher Musco, and Akash Rao. *Symposium on Discrete Algorithms (SODA) (to appear)*. 2026. [arXiv]

Fixed-sparsity matrix approximation from matrix-vector products

Noah Amsel, Tyler Chen, Feyza Duman Keles, Diana Halikias, Cameron Musco, and Christopher Musco. *SIAM Journal on Matrix Analysis and Applications (to appear)*. 2025. [arXiv]

A Unified Framework for Provably Efficient Algorithms to Estimate Shapley Values

Tyler Chen, Akshay Seshadri, Mattia J. Villani, Pradeep Niroula, Shouvanik Chakrabarti, Archan Ray,

Pranav Deshpande, Romina Yalovetzky, Marco Pistoia, and Niraj Kumar. *Conference on Neural Information Processing (NeurIPS) (to appear)*. 2025. [arXiv]

Optimal Polynomial Approximation to Rational Matrix Functions Using the Arnoldi Algorithm

Tyler Chen, Anne Greenbaum, and Natalie Wellen. *Numerical Algorithms*. 2025. [url][arXiv]

Randomized Matrix-Free Quadrature: Unified and Uniform Bounds for Stochastic Lanczos Quadrature and the Kernel Polynomial Method

Tyler Chen, Thomas Trogdon, and Shashanka Ubaru. *SIAM Journal on Scientific Computing*. 2025. [url][arXiv]

Near-optimal hierarchical matrix approximation from matrix-vector products

Tyler Chen, Feyza Duman Keles, Diana Halikias, Cameron Musco, Christopher Musco, and David Persson. *Symposium on Discrete Algorithms (SODA)*. 2025. [url][arXiv]

Near-Optimal Approximation of Matrix Functions by the Lanczos Method

Noah Amsel, Tyler Chen, Anne Greenbaum, Cameron Musco, and Christopher Musco. *Conference on Neural Information Processing (NeurIPS)*. 2024. [url][arXiv].

- invited for spotlight presentation.

Faster Randomized Partial Trace Estimation

Tyler Chen, Robert Chen, Kevin Li, Skai Nzeuton, Yilu Pan, and Yixin Wang. *SIAM Journal on Scientific Computing*. 2024. [url][arXiv]

Near-optimal convergence of the full orthogonalization method

Tyler Chen and Gérard Meurant. *ETNA - Electronic Transactions on Numerical Analysis*. 2024. [url][arXiv]

On the fast convergence of minibatch heavy ball momentum

Raghu Bollapragada, Tyler Chen, and Rachel Ward. *IMA Journal of Numerical Analysis*. 2024. [url][arXiv]

GMRES, pseudospectra, and Crouzeix's conjecture for shifted and scaled Ginibre matrices

Tyler Chen, Anne Greenbaum, and Thomas Trogdon. *Mathematics of Computation*. 2024. [url][arXiv]

A posteriori error bounds for the block-Lanczos method for matrix function approximation

Qichen Xu and Tyler Chen. *Numerical Algorithms*. 2024. [url][arXiv]

Stability of the Lanczos algorithm on matrices with regular spectral distributions

Tyler Chen and Thomas Trogdon. *Linear Algebra and its Applications*. 2024. [url][arXiv]

A spectrum adaptive kernel polynomial method

Tyler Chen. *The Journal of Chemical Physics*. 2023. [url][arXiv].

- This approach is implemented in the `spectral_density` package

Krylov-Aware Stochastic Trace Estimation

Tyler Chen and Eric Hallman. *SIAM Journal on Matrix Analysis and Applications*. 2023. [url][arXiv]

Low-Memory Krylov Subspace Methods for Optimal Rational Matrix Function Approximation

Tyler Chen, Anne Greenbaum, Cameron Musco, and Christopher Musco. *SIAM Journal on Matrix Analysis and Applications*. 2023. [url][arXiv]

Numerical computation of the equilibrium-reduced density matrix for strongly coupled open quantum systems

Tyler Chen and Yu-Chen Cheng. *The Journal of Chemical Physics*. 2022. [url] [arXiv]

Error Bounds for Lanczos-Based Matrix Function Approximation

Tyler Chen, Anne Greenbaum, Cameron Musco, and Christopher Musco. *SIAM Journal on Matrix Analysis and Applications*. 2022. [url] [arXiv]

Analysis of stochastic Lanczos quadrature for spectrum approximation

Tyler Chen, Thomas Trogdon, and Shashanka Ubaru. *International Conference on Machine Learning (ICML)*. 2021. [url] [arXiv].

- invited for long presentation

On the Convergence Rate of Variants of the Conjugate Gradient Algorithm in Finite Precision Arithmetic

Anne Greenbaum, Hexuan Liu, and Tyler Chen. *SIAM Journal on Scientific Computing*. 2021. [url] [arXiv]

Non-asymptotic moment bounds for random variables rounded to non-uniformly spaced sets

Tyler Chen. *Stat.* 2021. [url] [arXiv]

Predict-and-recompute conjugate gradient variants

Tyler Chen and Erin C. Carson. *SIAM Journal on Scientific Computing*. 2020. [url] [arXiv].

- abridged version was Student Paper Competition winner at 16th Copper Mountain Conference on Iterative Methods

Student Mentoring

Robert Chen (NYU)	2023 - 2025
Caroline Huber (NYU)	2024 - 2025
Ethan Lin (NYU)	2024 - 2025
Devin Tang (NYU)	2024 - 2025
Hajar Zaid (Graduate Center CUNY)	2024 - 2025
Ginebra Ferreira (NYU)	summer 2024
Kevin Li (NYU)	2022 - 2024
Yixin Wang (NYU)	2023 - 2024
Yue Geng (NYU)	summer/fall 2023
Ismael Jimenez (NYU)	summer 2023
Skai Nzeuton (Stuyvesant High School)	2022 - 2023
Yilu Pan (NYU Shanghai)	2022 - 2023
Qichen Xu (UW)	2021 - 2023
Linda Zhao (NYU)	summer/fall 2023
Aeron Langford (UW)	autumn 2019

Teaching

Instructor

Numerical Analysis (NYU MATH-UA 252)	fall 2024
Linear Algebra I (NYU MATH-GA 2110)	spring 2024
Numerical Analysis (NYU MATH-UA 252)	fall 2023
Numerical Analysis (NYU MATH-UA 252)	spring 2023
Mathematical Statistics (NYU MATH-UA 234)	fall 2022
Applied Linear Algebra and Numerical Analysis (UW AMATH 352)	spring 2021
Interdisciplinary Writing/Natural Science (UW ENGL 199)	winter 2021
Interdisciplinary Writing/Natural Science (UW ENGL 199)	autumn 2020

TA or Grader

Probability and Statistics for Computational Finance, TA (UW CFRM 410)	winter 2019
Calculus with Analytic Geometry I, TA (UW MATH 124)	autumn 2018
Calculus with Analytic Geometry II, TA (UW MATH 12)	winter 2018
Calculus with Analytic Geometry II, TA (UW MATH 125)	autumn 2017
Electronics, TA (Tufts PHY 41)	spring 2017
Electronics, TA (Tufts PHY 41)	spring 2016
Discrete Mathematics, Grader (Tufts MATH 61)	spring 2016
Calculus III, Grader (Tufts MATH 42)	fall 2015
Differential Equations, Grader (Tufts MATH 51)	spring 2015
Calculus III, Grader (Tufts MATH 42)	fall 2014

Talks and Posters

Householder XXII, Presentation [pdf]	2025
Joint Math Meetings, Presentation [pdf]	2025
Mid-Atlantic Numerical Analysis Day, Presentation [pdf]	2024
NYU Theory Seminar, Presentation [pdf]	2024
Precond, Presentation [pdf]	2024
SIAM Linear Algebra, Presentation [pdf]	2024
Georgia Tech, Presentation [pdf]	2024
NYU Math Modeling workshop, Presentation	2023
NYU Math Society meeting, Presentation	2023
SIAM New York, New Jersey, and Pennsylvania Annual Meeting, Presentation [pdf]	2023
Universität Bielefeld, Presentation [pdf]	2023
NYU Shanghai, Presentation [pdf]	2023
International Congress on Industrial and Applied Mathematics, Presentation [pdf]	2023

Sampling Theory and Applications Conference, Presentation [pdf]	2023
Foundations of Computational Mathematics, Presentation [pdf]	2023
Perspectives on Matrix Computations, Presentation [pdf]	2023
Courant Numerical Analysis and Scientific Computing Seminar, Presentation [pdf]	2022
Conference on Random Matrix Theory and Numerical Linear Algebra, Presentation [pdf]	2022
Workshop on Algorithms for Large Data (Online), Poster [pdf]	2021
International Conference on Machine Learning, Oral	2021
SIAM Linear Algebra 21, Presentation [pdf]	2021
Baidu Research, Presentation [pdf]	2021
Householder Symposium, Poster (Cancelled)	2020
Copper Mountain Student Paper Award Session, Presentation (Cancelled)	2020
SIAM Parallel Processing, Presentation [pdf]	2020
Baidu Research, Presentation	2019

Service and Outreach

Proud to Be First Faculty Connect	2023-2024
Serve as mentor for Proud to Be First Faculty Connect, which pairs second-year, first-generation students with faculty	
Math Modeling Workshop	nov. 2023
Provide tutorial on randomized linear algebra for the math modeling club	
Math Society chalkboard talk	nov. 2023
Guest lecture for NYU Math Society on randomized linear algebra	
NYU SIAM podcast	oct. 2023
Discuss my path as a mathematician, and advice for students, etc.	
NYU SIAM Grad School Info Session	oct. 2022
Panelist for Q/A session for students interested in grad school	
Minisymposium Organizer	may 2021
Random matrices and numerical linear algebra (at SIAM Linear Algebra 21, co-organized with Thomas Trogdon) [program]	
Graduate Student Representative	2019 - 2020
Represent interests of graduate students to the department	
Minisymposium Organizer	feb. 2020
High performance Krylov subspace methods: Theory, Implementation, and Application (at SIAM Parallel Processing 20) [program]	
Diversity Committee Departmental Climate Orientation	oct. 2019
Panelist for event focused on building an inclusive department culture	

Numerical Analysis Research Club 2019 - 2020
Organize and plan weekly meetings for NARC

SIAM UW Mental Health Conversation and Resources oct. 2018
Organize and facilitate a discussion about mental health in grad school

Software

Research code (<https://github.com/tchen-research>)

Repositories with code to generate figures and experiments from my papers.

Spectral Density (<https://pypi.org/project/spectral-density/>)

Develop `spectral_density` package to efficiently produce spectrum adaptive KPM approximations.

PETSc (<https://www.mcs.anl.gov/petsc/>)

Contribute PIPEPRCG. This method can be used by with the flag `-ksp_type pipeprcg`.

mpmath (<https://github.com/mpmath>)

Update matrix multiplication driver to significantly improve performance for sparse matrices.

Awards & Honors

Boeing Research Award (UW Department of Applied Mathematics) 2020

Student Paper Competition Winner (Copper Mountain Conference on Iterative Methods) 2020

Graduate Research Fellowship (NSF) 2019

Top Scholars Fellowship (UW) 2017

The Audrey Butvay Gruss Science Award (Tufts) 2017

Phi Beta Kappa (Tufts) 2017

Sigma Pi Sigma Physics Honors Society (Tufts) 2016

The Howard Sample Prize Scholarship in Physics (Tufts) 2015